

Project Description: Predict Optimal Operational Expenditure with Linear Surrogate Model

Motivation:

Investment decisions for industrial multi energy systems (MES) are often supported by Mixed Integer Linear Program (MILP) models. Solving these MILPs can be computationally demanding – especially when uncertainties are addressed through *post-optimality analysis* such as Monte Carlo Analysis, which require repeated model evaluations. In such cases, surrogate models can be used to emulate the input-output relation of the MILP (see *Figure 1*), significantly reducing computational effort. Surrogate models are trained on a limited set of data points (i.e., training set) to learn the behavior of a complex, expensive to evaluate model. Surrogate model types range from simple linear regressions to more complex architectures like artificial neural networks [1]. Once trained, surrogate models allow for rapid prediction of expensive to evaluate models.

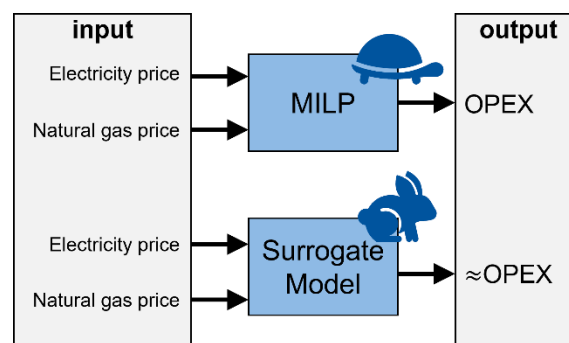


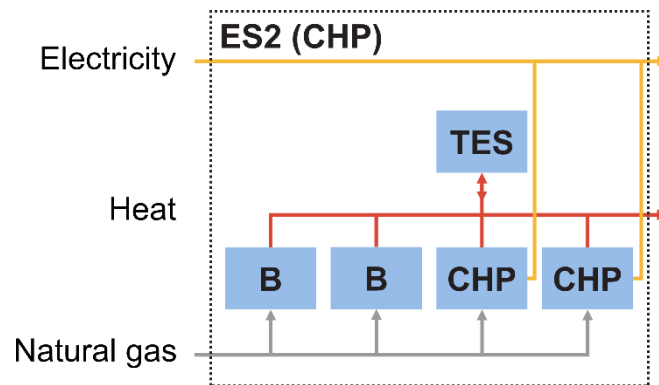
Figure 1: Input-output relation of a MILP and a surrogate model. In this project, the inputs are average electricity prices and natural gas prices. The output is the operational expenditure (OPEX).

Project Objective:

Your task is to fit a linear regression model to predict the optimal operational expenditure (OPEX) of an MES design depending on the energy carrier prices. Use the MES design “ES2 (CHP)” from [2]. The objectives are as follows:

- **Model Generation:** Model a weekly operational optimization as MILP to determine the OPEX of the MES design. Create a second model where you simplify the MILP into a Linear Program (LP).
- Define realistic price ranges for electricity and natural gas.
- **Sample Generation:** Generate a training set and a test set of random energy carrier price scenarios and determine the optimal OPEX for each individual scenario with the MILP. The training set should contain 40 samples. The test set should contain 10 samples.
- **Surrogate Modeling:** Fit a linear regression (LR-MILP) model to predict OPEX with the training set. Utilize the test set to analyze the prediction quality using the Mean Squared Error and the Coefficient of Determination R^2 .
- Fit another linear regression (LR-LP) to a training set generated with the LP model. How do the prediction qualities of LR-MILP and LR-LP differ?
- Fit both linear regressions (LR-MILP and LR-LP) with smaller training sets (use 5 samples and 20 samples). How does the size of the trainings sets influence the prediction quality of LR-MILP and LR-LP?

Industrial Multi Energy System:



Data:

- Technology data for combined heat and power (CHP) units, natural gas boilers (B), and thermal energy storage (TES) can be found in [2].
- Time-dependent energy demands can be found in the *energy_demands.csv*.
- Price data for industrial consumers in Germany:
 - Natural gas (historic): <https://www.statista.com/statistics/1346785/gas-prices-development-commercial-industrial-customers-germany/>
 - Electricity prices (historic): <https://www.statista.com/statistics/1050448/industrial-electricity-prices-including-tax-germany/>

Assumptions:

- MILP: For the natural gas boilers and combined heat and power (CHP) units, model the minimal part load and the part-load behavior.
- Use time-independent natural gas and electricity prices.
- The linear regression models each have two inputs (electricity price and natural gas price) and one output (predicted OPEX).

Sources:

[1]: Forootan et al. (2022), Machine Learning and Deep Learning in Energy Systems: A Review

[2]: Seele et al. (2023), Optimization-based energy system planning under long-term uncertainty: Rapid Monte Carlo analysis using linear regression